CAREPULSE OPD SUITE

CAPSTONE 2 DEFENSE

FULL-STACK OUTPATIENT CLINIC ARCHITECTURE

Multi-Doctor Outpatient Clinic Management System

End-to-end departmental synchronization across Patient Registration, Live Token Queuing, Digital Prescriptions, and Pharmacy Billing powered by Supabase.

TIMELINE
3-Week Sprint

BACKEND ENGINE
Supabase PostgreSQL + RLS

PRESENTERS
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01 / EXECUTIVE OVERVIEW

Operational Bottlenecks vs. Integrated Solution

Fragmented Clinic Workflows

Manual Queuing: Long waiting room delays and overcrowded reception desks.

Paper Prescriptions: Illegible handwriting causing pharmacy dosage errors.

Siloed Records: Cashiers unable to access accurate itemized medicine billing.

CarePulse Integrated Ecosystem

Real-time Tokens: Automated daily token creation (T-001) per doctor queue.

Digital Rx Repeater: Instant line-item drug selection with allergy safety checks.

Unified Cashier Hub: Auto-calculated medicine pricing, tax, and printed slips.

Goal: Eliminate manual paper trails and reduce patient wait times by 60%.

02 / ARCHITECTURE

Interdepartmental Patient Lifecycle

1

Reception Desk

MRN creation & daily doctor queue token assignment.

→

2

Live Queue Display

Real-time token calling with audio chime alerts.

→

3

Doctor Desk

Vitals logging & digital line-item prescription builder.

→

4

Pharmacy & Billing

Inventory check, invoice calculation & printed receipt.

Core Concept Practised: Shared AngularJS Services + Supabase Realtime State Synchronization across four independent views.

03 / DATA LAYER

Relational Schema & Entity Lifecycles

patients

Primary demographics record with unique MRN identifier key.

visits

Tracks token numbers, live doctor queues, symptoms, and vitals.

prescriptions

Stores dynamic line-item drug repeaters inside JSONB columns.

invoices

Stores consultation fees, medicine total, tax, and payment modes.

```
$scope.prescription = { patientId: "MRN-8841", doctor: "Dr. A. Rao", medicines: [{ name: "Amoxicillin", dosage: "1-0-1" }]
};
```

JSONB Storage Format

04 / FRONT-OFFICE OPERATIONS

Patient Check-in & Waiting Room Display

Reception Registration

Automated MRN Generation: Assigns unique patient keys instantly.

Duplicate Detection: Flags phone number collisions during check-in.

Daily Token Issuer: Sequential daily token format (e.g., T-014).

Live Queue Screen

Doctor Queue Split: Isolated token screens per consulting room.

Call Next Control: One-click token progression with audio chimes.

State Sync: Live status markers (Waiting, In Consultation, Complete).

TOKEN FORMAT	AVERAGE WAIT TIME	SYNC LATENCY
T-001 to T-999	< 12 Mins	Realtime (< 200ms)

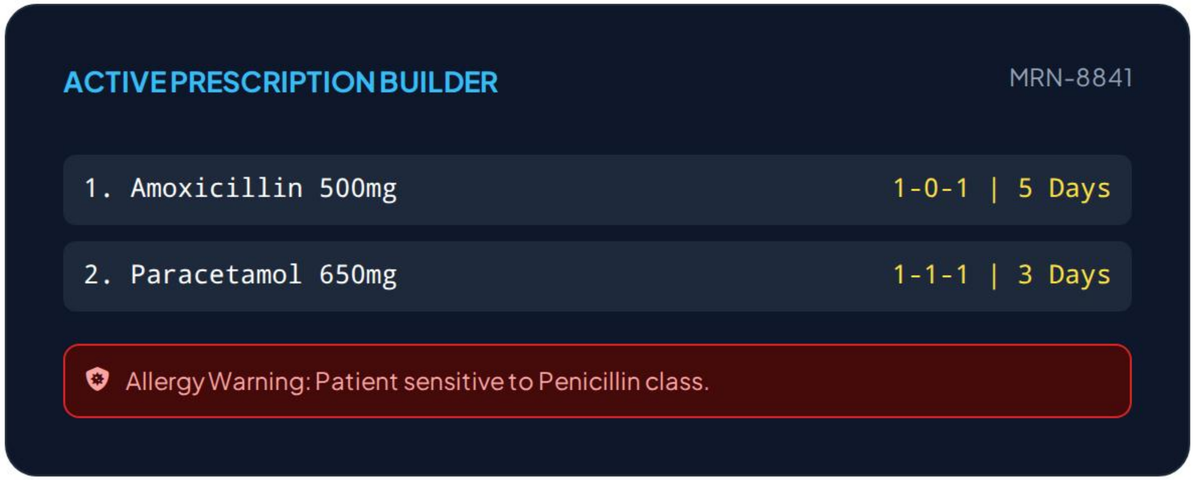
05 / CLINICAL OPERATIONS

Doctor Desk & Dynamic Prescription Builder

Clinical Examination Suite

Physicians record vitals (BP, Pulse, Temp, Weight), chief complaints, and clinical diagnoses while having immediate access to historical visit records.

- **Dynamic Repeater:** Add/remove medicine rows seamlessly.
- **Dosage Controls:** Structured timing flags (1–0–1, 1–1–1, SOS).
- **Safety Safeguards:** Automated allergy warning popups.



Output: Completed consultations automatically route prescriptions to the Pharmacy Module.

06 / PHARMACY & CASHIER

Dispensing, Invoice Calculation & Slips

ITEM DESCRIPTION	CATEGORY	QTY / DAYS	UNIT PRICE	SUBTOTAL
Doctor OPD Consultation Fee	Clinical Service	1 Session	₹ 500.00	**₹ 500.00**
Amoxicillin 500mg Capsules	Pharmacy Line Item	10 Caps	₹ 12.00	**₹ 120.00**
GST / Clinic Tax (5%)	Statutory Tax	5% Standard	-	**₹ 31.00**

Printable Header Slips **Multi-Mode (Cash / UPI / Card)**

NET GRAND TOTAL: ₹ 651.00

07 / PATIENT HEALTH RECORDS

Historical Visit Search & EHR Lookup

Multi-Parameter Lookup

Instantly query complete medical records by entering either the Patient Phone Number or unique Medical Record Number (MRN).

Search: "MRN-8841" or "9876543210"

OCT 14, 2026 • VISIT #04 — **Dr. Ananya Rao**

Diagnosis: Acute Bronchitis

Meds: Amoxicillin 500mg, Cetirizine 10mg

MAY 02, 2026 • VISIT #01 — **Dr. Vikram Seth**

Diagnosis: Seasonal Rhinitis

Meds: Montelukast 10mg

Advantage: Complete longitudinal patient history accessible across all four department workstations.

08 / REPORTS & PERFORMANCE

Daily OPD Analytics & Technical Highlights

TOTAL DAILY REVENUE

₹42,850

VISIT QUEUE VOLUME

68 Tokens

AUTOMATIC MOCK FALLBACK

100% Zero-Fail

AngularJS ngRoute

Supabase PostgreSQL

Tailwind CSS

Row Level Security (RLS)



Capstone Project Complete

CarePulse Multi-Doctor OPD System successfully demonstrates scalable, multi-departmental healthcare workflows with real-time database synchronization.

Thank You — Questions & Answers